Spring 2021 Real Analysis II — Homework 6

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Unless otherwise specified, $C([a,b])$ is endowed with norm $\|x\| = \max_{a\le t\le b} |x(t)|$ for any $x \in C([a,b])$.

1. Let X be a B* space and X_0 a closed linear subspace. Show that $d(x, X_0) = \sup\{|f(x)| : f \in X^*,\ \|f\| = 1,\ f|_{X_0} = 0\}$.

2. Let X be a B* space. Suppose there are n linearly independent vectors $\{x_i : i = 1, \dots, n\}$ in X, n constants $\{c_i : i = 1, \dots, n\} \subset \mathbb{R}$, and $M > 0$. Show that the following two statements are equivalent:
   - There exists $f \in X^*$ such that $f(x_i) = c_i$ for all i and $\|f\| \le M$.
   - For any $a_i \in \mathbb{R}$ there is $|\sum_{i=1}^n a_i c_i| \le M \|\sum_{i=1}^n a_i x_i\|$.

3. Let X be a B* space and $x_1, \cdots, x_n$ be independent vectors in X. Show that there exist $f_1, \dots, f_n \in X^*$ such that $f_i(x_j) = \delta_{ij}$ for all $i, j = 1, \dots, n$.

4. Let $C_0 := \{\xi = (\xi_1, \xi_2, \dots) : \xi_k \in \mathbb{R},\ \lim_{k\to\infty} |\xi_k| = 0\}$ and define its norm $\|\xi\| := \sup_{k\ge 1} |\xi_k|$. Show that $C_0^* = l_1$.

5. Let $x_k, x \in C([a,b])$ for all $k \in \mathbb{N}$ and $x_k \rightharpoonup x$. Show that $\lim_{k\to\infty} x_k(t) = x(t)$ for all $t \in [a,b]$, i.e., $x_k \to x$ pointwisely.

6. Let X be a B* space and $x_k \rightharpoonup x_0$. Show that $\liminf_{k\to\infty} \|x_k\| \ge \|x_0\|$.

7. Let X be a separable Hilbert space and $\{e_k : k \in \mathbb{N}\}$ a basis (complete orthonormal set) of X. Show that $x_n \rightharpoonup x_0$ if and only if both of the following statements hold:
   - There exists $M > 0$ such that $\|x_n\| \le M$ for all $n \in \mathbb{N}$.
   - For any $k \in \mathbb{N}$, there is $\lim_{n\to\infty} \langle x_n, e_k \rangle = \langle x_0, e_k \rangle$.

8. Let $p \in (1, \infty)$ and $X = L^p(\mathbb{R})$ be endowed with the L^p norm $\|\cdot\|_p$. Define $T_k \in L(X)$ by $(T_k x)(t) = x(t+k)$ for any $t \in \mathbb{R}$, $x \in X$, and $k \in \mathbb{N}$. Show that $T_k \rightharpoonup 0$ but $\|T_k x\|_p = \|x\|_p$ for all $k \in \mathbb{N}$. [Hint: for any $\epsilon > 0$ and $v \in L^q(\mathbb{R})$, show that both $\int_{-\infty}^{-a} v(t)u(t+k)\,\mathrm{d}t$ and $\int_{-a}^{\infty} v(t)u(t+k)\,\mathrm{d}t$ can be bounded by $\epsilon/2$ for sufficiently large a and k.]

9. Let X be a Hilbert space, $x_k \rightharpoonup x_0$, and $y_k \to y_0$. Show that $\langle x_k, y_k \rangle \to \langle x_0, y_0 \rangle$.

10. Let X be a B* space. Show that any closed convex set E in X is weakly closed. (E is called *weakly closed* if $x_0 \in E$ whenever $\{x_k\} \subset E$ and $x_k \rightharpoonup x_0$.)